

A Transparent Screening Framework for Preliminary Engineering Investigation Prioritization of Mine-Waste Features Using the 2026 USGS National Mine Waste Inventory

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Abstract

A transparent desktop screening framework was developed to prioritize mine-waste features for preliminary engineering investigation. This framework is applied when the investigation resources are limited, and 760 valid mine-waste features in the 2026 USGS National Mine Waste Inventory were classified. The inventory resource records were used to calculate feature-level volume, transformed using a base-10 logarithm, and min-max normalization was used to produce a volume score between 0 and 1. Based on this volume score, three bands were created based on tertile thresholds: Higher, Intermediate, and Lower. Four domains, material, geochemical, geotechnical, and water/containment, were considered to develop engineering investigation profiles based on feature-type-level engineering evidence. Two main problems were addressed: which feature should be considered first based on their physical scale, and which engineering aspect should be considered during subsequent investigation. The consistency of priority-band assignments, engineering-profile classifications, final output records, boundary identification, and reproducibility were confirmed during validation, and the framework can be reproduced within a predefined numerical tolerance. This framework is considered a preliminary desktop screening tool rather than a specific investigation urgency, such as environmental risk, economic value, resource recovery potential, or any other factor. This provides a structured basis for selecting a mine-waste feature for more detailed investigation.

1. Introduction

Extensive quantities of different types of waste materials are generated due to mining and mineral processing activities. Most of these waste materials, such as tailings and waste rock, are stored at or near mine sites for a longer period. Depending on the type of the mineral deposit, particle size distribution, processing methods used, mineralogy, and storage methods, the physical and chemical characteristics of these materials vary. The U.S. Geological Survey (USGS) National Mine Waste Inventory is a geospatial database that provides information about surface mine waste features associated with both historical and present mining activities. Information such as location details, size, commodity, and mineralogy of features is included which is gathered from publicly available sources by the USGS and participating State geological surveys (*National Mine Waste Inventory | U.S. Geological Survey*, n.d.).

Mine-waste features have considerable differences in their physical and chemical behavior. Due to these different properties, these features cannot be considered a single homogeneous group. As an example, mine tailings are fine residual materials generated during mineral processing and may contain residual sulfide minerals and constituents related to processing. These properties can particularly influence their storage behavior, geochemical reactions, and interaction with the surrounding environment, which later influences long-term management (Edraki et al., 2014).

When consider waste-rock and overburden features, their material characteristics and interaction with surrounding environment should be required to consider. The literature mentions that these behaviors can result in coupled geochemical, mineralogical, and physical transport processes, such

as waste and gas movement through waste-rock piles. The drainage quality and quantity of piles can vary based on material properties and site conditions (Vriens et al., 2020).

These features may require preliminary investigation based on several engineering domains such as material characteristics, geochemical behavior, geotechnical characteristics, and water/containment related conditions. The inventory does not provide a direct investigation structure. The inventory contains documented details about individual features. For a detailed engineering investigation, it requires more site-specific information. Therefore, a transparent screening process is required to organize these features according to their relevance for preliminary engineering investigation.

This study developed a preliminary screening framework that addresses two separate questions: which feature should be considered for preliminary investigation earlier, and which engineering domain should be considered during the investigation of the specific feature? Features considered for preliminary investigation earlier are selected based on a physical scale represented by their calculated volume. Investigation profiles were developed according to the waste classification included in the dataset to decide which engineering domain should be considered. The inventory is used as a basis for a preliminary screening framework, preserving the original available site information.

A normalized Volume Score is assigned to each valid feature and categorized into three Priority Bands: Higher, Intermediate, and Lower screening categories. Simultaneously, they are assigned to investigation profiles based on their engineering domain relevance: material, geochemical, geotechnical, and water/containment. The framework is applied to 760 valid mine waste records from the inventory. The resulting framework is a transparent and reproducible screening procedure for preliminary investigation work without specifying an objective.

This cannot be considered a direct measurement of risk or other engineering performance, as those strongly depend on site-specific conditions. This study supports decisions on further investigations such as field investigation, sampling, testing, and engineering analysis.

2. Problem Statement

The 2026 USGS National Mine Waste Inventory consists of information on documented mine-waste features with different waste types, volumes, and characteristics. This inventory does not provide a direct, transparent engineering structure that supports the initiation of preliminary investigation. The information available in the database is not sufficient to consider site-specific conditions of features such as geotechnical stability or groundwater behavior.

Therefore, it is required to develop a preliminary screening framework that organizes the features included in the database according to a relative prioritization order, with the identification of their engineering domains that require subsequent investigation.

3. Research Objective

To develop a transparent and reproducible screening framework for prioritizing mine-waste features for preliminary engineering investigation using information available in the 2026 USGS National Mine Waste Inventory.

- To develop a normalized feature-level Volume Score.
- To classify features into preliminary Priority Bands from the calculated volume data.
- To develop engineering investigation profiles that identify the material, geochemical, geotechnical, and water/containment domains involving consideration.
- To evaluate and validate the consistency, reproducibility and methodological logic of the framework,

4. Dataset and Study Population

The study uses the 2026 USGS National Mine Waste Inventory, which contains documented information about surface mine waste features. It contains information on mine-waste features such as feature location, waste type, volume, geological information, commodity, and other related information (*National Mine Waste Inventory* | *U.S. Geological Survey*, n.d.) .

There is information about 764 mine-waste features, which are represented using waste points and spatial polygon datasets. Each feature consists of its resource-related and geological information with site identifiers. As the study is based on volume data for the primary screening methodology, 760 features with valid positive volume were selected.

Therefore, the final dataset contains information about 760 features and is used for the proposed screening framework. Features with missing volume records and negative volume records were excluded from the volume-based screening analysis.

Table 5.1. USGS Inventory Layers Used in the Study

Layer	Purpose
Waste_Points	Feature identification and primary feature attributes
Waste_Resources	Resource/volume information
Waste_Geology	Deposit type, commodity and mineralogical context
Waste_Polygons	Spatial extent / geometry
Waste_References	Supporting source references

5. Data Preparation and Quality Control

Detailed analysis was carried out in order to identify the missing data, inconsistent records, and variables without sufficient information. Using available feature and site identifiers, relationships between each feature, resource, and their geological data were checked.

Feature-level calculated volume data were inspected, and the median of the calculated volumes of multiple resources having the same feature was used to obtain a representative volume for the specific feature. As the screening was volume-based, negative and missing volume records were excluded, resulting in 760 feature records.

The other attributes in the inventory were used for descriptive analysis, engineering classification, and contextual interpretation. Variables with substantial completeness in the data were used as primary screening components in the framework, as this avoided introducing assumptions or values for the inventory.

6. Framework Methodology

The framework produces two separate outputs:

1. Priority Band – determines the relative order based on their physical scale where a feature may be considered for preliminary investigation.
2. Investigation Profile – identifies the engineering domains that require consideration during subsequent investigation.

The calculated volume is used to construct the Priority Band based on the relative physical scale of features. Four engineering domains- Material, Geochemical, Geotechnical, and Water/Containment- were considered, and features were classified into categorical levels. These two outputs are independent of each other.

7. Volume-Based Screening Methodology

7.1. Feature-Level Volume Calculation

The calculated volume (Calc_Vol) data are used as the measure of the physical scale of mine-waste features. The median was calculated to obtain a representative volume for each feature ID (Ftr_ID) when a feature is associated with many resource records. This reduces the influence of extremely large or small values when there are multiple records.

From the excluded 4 features, three features had negative calculated volumes, and one feature did not have a calculated volume. As they were not realistic data for the screening framework, those 4 feature records were excluded. 760 valid feature records were selected as the final dataset for the analysis.

7.2. Logarithmic Transformation

The volume distribution of the final selected dataset was extremely right-skewed. The differences between the physical scales of features were substantially large. Therefore, a base-10 logarithmic transformation was applied to the dataset.

$$V_{log} = \log_{10} V$$

where V is the calculated feature-level volume.

This transformation solved the problem of extremely large volume values and provided a more suitable distribution for the dataset.

7.3. Normalized Volume Score

The normalized score ranging from 0-1 was built up using transformed volumes using min-max normalization:

$$Volume\ Score = \frac{V_{log} - V_{log,min}}{V_{log,max} - V_{log,min}}$$

According to the normalization, features closer to 1 represent features with larger volume, while smaller-volume features are closer to 0 when compared to other features in the population.

This volume score was considered only as a screening-scale indicator, and it is not interpreted as a measure of any specific investigation urgency.

7.4. Priority-Band Construction

Three screening bands, which are relative within the population studied, were established using the normalized Volume Score. Two classification thresholds are the one-third and two-thirds quantiles of the Volume Score distribution. The 760 features were divided into three preliminary screening priority groups: Higher, Intermediate, and Lower.

The Higher and Intermediate Priority Bands contain 253 features each, while the Lower Priority Band contains 254 features, where the resulting thresholds were 0.4136200090391657 and 0.6602075879185684.

The upper third of the Volume Score distribution is represented by the Higher band, the middle portion is represented by the Intermediate band, and the remaining/lower portion of the Volume Score distribution is represented by the Lower band. In the final framework, these bands represent the relative order of features for preliminary investigation based on their physical scale.

This is not a measure of any specific investigation urgency, such as environmental risk, resource recovery, or remediation urgency.

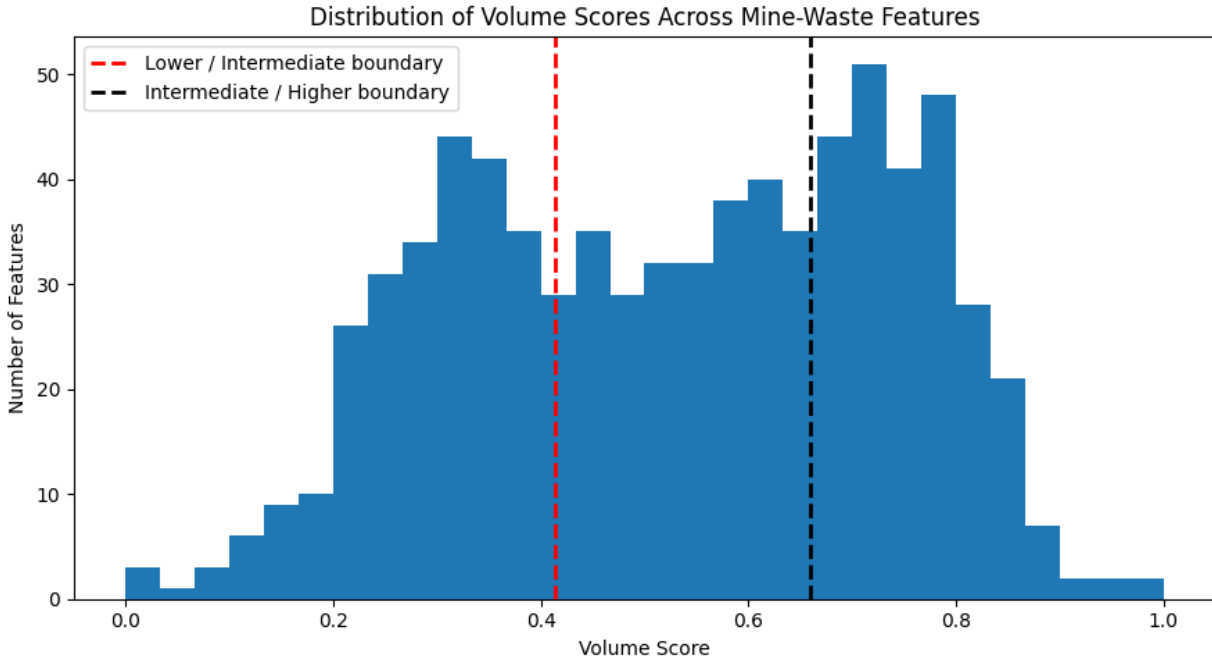


Figure 7.1. Distribution of Volume Scores Across Mine-Waste Features

7.5. Engineering Investigation Profiles

Engineering investigation profile system was developed including four main engineering domains which may require to consider during subsequent investigation. The four domains were:

1. Material
2. Geochemical
3. Geotechnical
4. Water/Containment

Referring to the documented waste-feature classification, categorical levels from 0 to 2 were assigned to each domain.

Table 7.1. Categorical Level Definitions

Level	Meaning
0	The available feature classification evidence is not sufficient to assign engineering consideration.
1	The basis for engineering investigation consideration is provided, but the characteristics are not sufficient to define or assign level 2.
2	Strong evidence is available to consider explicitly during subsequent investigation.

These levels are considered only categorical levels rather than numerical weights. Therefore, they do not provide any numerical definition.

Six standard investigation profiles were produced using a combination of four domains.

Table 7.2. Engineering Investigation Profile Matrix

Profile	Material	Geochemical	Geotechnical	Water/Containment
P1	2	2	2	1
P2	2	2	1	2
P3	1	2	1	2
P4	2	2	1	0
P5	2	1	1	0
P6	0	0	0	0

Section 8 includes the engineering reasoning and literature support for these assignments.

7.6. Contextual Variables

There are several attributes that are considered contextual variables and were not used in the Volume Score and Priority Band calculation. Reclaimed status, Superfund status, Commodity, Commodity Count, Boundary Distance, and Boundary Flag are the attributes that were used as contextual variables.

Reclaimed and Superfund status information were considered as management or regulatory context for each feature. However, these variables were not taken into the major engineering priority measurement. On the other hand, reclaimed status does not provide solid evidence that a feature is stable or requires no further investigation. Same as reclaimed status, Superfund status also does not provide evidence for higher engineering investigation priority.

Geological and mining-related information were provided by commodity information. Commodity count is used as a descriptive representative rather than a measure of any specific investigation urgency.

Therefore, for the contextual interpretation and for the subsequent investigation decision-making, these attributes were kept rather than being considered as additional scoring components. This prevents changing the primary volume-based screening order and avoids redundancy.

7.7. Boundary Cases

The Priority Band thresholds are derived from the Volume Score distribution. As a result of small changes in the fundamental volume calculation or numerical representation, the features closer to a threshold may change bands.

To identify those scenarios, the absolute distance between each feature's Volume Score and the nearest priority-band threshold was calculated as:

$$D_i = \min(|S_i - T_1|, |S_i - T_2|)$$

where S_i is the Volume Score of feature i and T_1 and T_2 are the two priority-band thresholds.

When this distance is less than 0.02, a Boundary Flag was assigned. This resulted in 76 boundary cases within the population.

This Boundary Flag identifies features when their position is closer to a screening threshold, and it provides a warrant indicating an additional attention requirement when interpreting the band assignment.

8. Engineering Investigation Profiles

Each mine-waste feature group is categorized into engineering investigation profiles by assigning a categorical level for Material, Geochemical, Geotechnical, and Water/Containment considerations. During the assignment, general characteristics of each mine-waste feature and published categorization approaches were considered. As mentioned in the literature, mine waste categories such as tailings, waste rock, heap-leach piles, ore stockpiles, slag, and other processed material are identified and categorized separately from the mine-water sources (Campbell et al., n.d.).

8.1. P1 – Tailings and Rock/Overburden

Tailings and rock/overburden features were assigned P1: Material 2, Geochemical 2, Geotechnical 2, and Water/Containment 1.

Tailings are the residues produced during processing activities with physical and chemical characteristics that can affect several factors, such as storage behavior, geochemical reactions, and water contamination. According to the USGS characterization, tailings are classified as fine-grained mine-waste materials which require specific sampling and characterization (Campbell et al., n.d.). Studies showed that these tailings and rock/overburden can be subjected to coupled geochemical and physical transport, resulting in geochemical and water interactions (Vriens et al., 2020).

Because of the physical properties, mineralogical composition, and interaction with water, rock/overburden features have to be considered mainly in material, geochemical, and geotechnical domains during subsequent investigations. These rock piles may exhibit substantial physical heterogeneity (Vriens et al., 2020).

According to the available information on tailings and rock/overburden features, this does not imply that every feature consists of a water/containment structure. Therefore, the water/containment consideration is much less than the other three domains. However, it remains a subsequent investigation consideration, but the inventory does not establish strong containment characteristics.

8.2. P2 – Leach-Related Features

Leach-related features were assigned P2: Material 2, Geochemical 2, Geotechnical 1, and Water/Containment 2.

Heap-leach piles are identified as a different mine-waste type in the USGS characterization protocol (Campbell et al., n.d.). During the leaching process, stored material interacts with an applied or infiltrating solution. As a result, there is a direct relevance to the subsequent investigation in geochemical behavior and water/containment management.

As the stored material is the basis for the leaching process, material was assigned level 2. Due to the above-mentioned interaction, Geochemical and Water/Containment domains were assigned level 2. Geotechnical considerations are relevant but not sufficient to characterize each leach-related feature as a strong consideration.

8.3. P3 – Water / Process Ponds

Water/process ponds were assigned P3: Material 1, Geochemical 2, Geotechnical 1 and Water/Containment 2.

USGS mine-waste characterization protocols consider mine-water sources as a separate component, and further investigation includes water sampling, field measurements, and flow-related observations (Campbell et al., n.d.).

As water itself is a defining characteristic, Water/Containment was assigned level 2. Stored water/process ponds may require subsequent investigation on chemical characterization. Therefore, level 2 was assigned to the Geochemical domain. As there is a lack of information on material and structural conditions, the Material and Geotechnical domains were assigned level 1.

8.4. P4 – Process / Metallurgical Residues

Process/metallurgical residues were assigned P4: Material 2, Geochemical 2, Geotechnical 1, Water/Containment 0.

Material and geochemical domains were assigned level 2 because the material characteristics of process residue are related to the subsequent characterization and composition, and the mineralogical characteristics of the residue can influence the constituent release. Physical behavior is relevant but not sufficient to define characteristics from the classification alone.

Evidence for water/containment is not sufficient to determine whether these residues are water-containment features. Therefore, level 0 was assigned. But it does not imply that water interaction is not relevant for a feature.

8.5. P5 – Ore / Mineralized Stockpiles

Ore/mineralized stockpiles were assigned P5: Material 2, Geochemical 1, Geotechnical 1, and Water/Containment 0.

Stored materials themselves are a defining feature. Therefore, the material was assigned level 2. When considering the waste-feature group, it does not support the relevance of geochemical and geotechnical to level 2. But in subsequent investigations, these two domains may be relevant. It also does not provide any evidence of a water-related or containment function. Therefore, water/containment was assigned level 0.

8.6. P6 – Other / Unclassified

Other/unclassified features were assigned P6: Material 0, Geochemical 0, Geotechnical 0, and Water/Containment 0.

These features do not provide sufficient information related to the four domains. Level 0 does not imply that these features do not require a subsequent investigation into a specific domain, but it is not sufficient to assign a generalized engineering profile.

9. Results

9.1. Priority-Band Distribution

Three preliminary Priority Bands were produced when the 760 valid features were assigned to tertile thresholds based on their Volume-Score.

Table 9.1. Priority-Band Distribution

Priority Band	Features	Percentage
Higher	253	33.3%
Intermediate	253	33.3%
Lower	254	33.4%
Total	760	100%

The Volume Score thresholds were 0.4136200090391657 and 0.6602075879185684.

9.2. Engineering-Group Distribution Across Priority Bands

The distribution of engineering groups across the three Priority Bands was:

Table 9.2. Engineering Groups with Priority Bands

Engineering Group	Higher	Intermediate	Lower
Leach-related	60	25	4
Ore / mineralized stockpile	0	4	3
Other / unclassified	0	0	1
Process / metallurgical residue	0	10	4
Rock / overburden	82	72	101
Tailings	101	137	138
Water / process pond	10	5	3

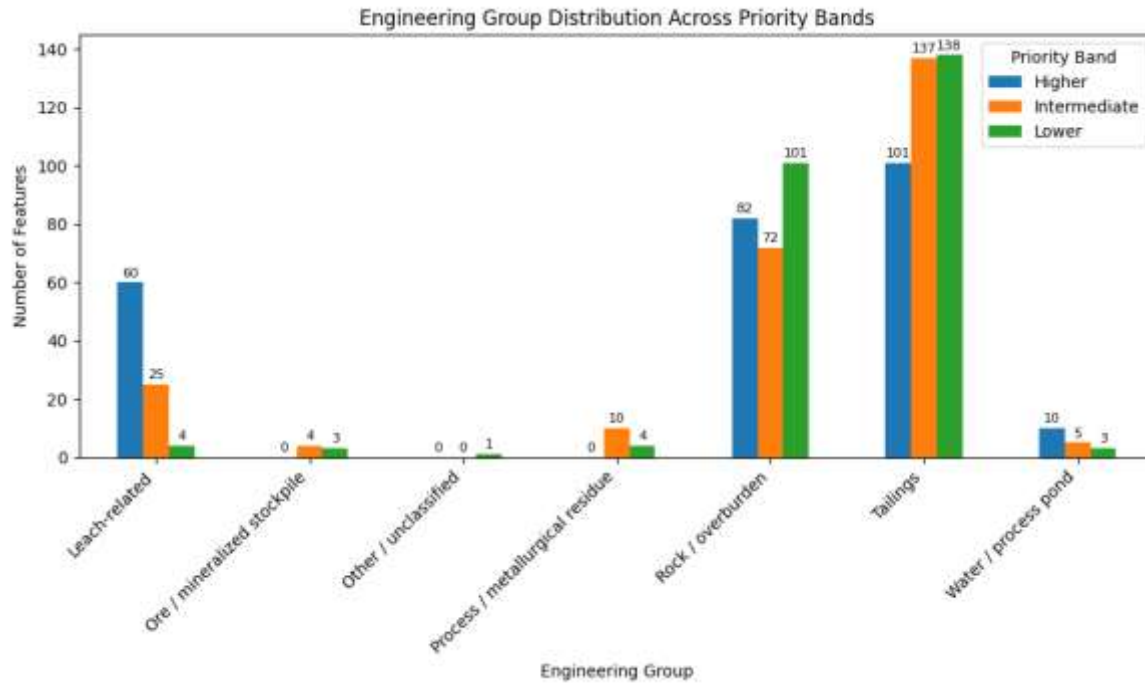


Figure 9.1. Engineering Group Distribution Across Priority Bands

These observations are due to the distribution of Volume Scores within the groups and not due to any assignment of numerical priority weights to the groups.

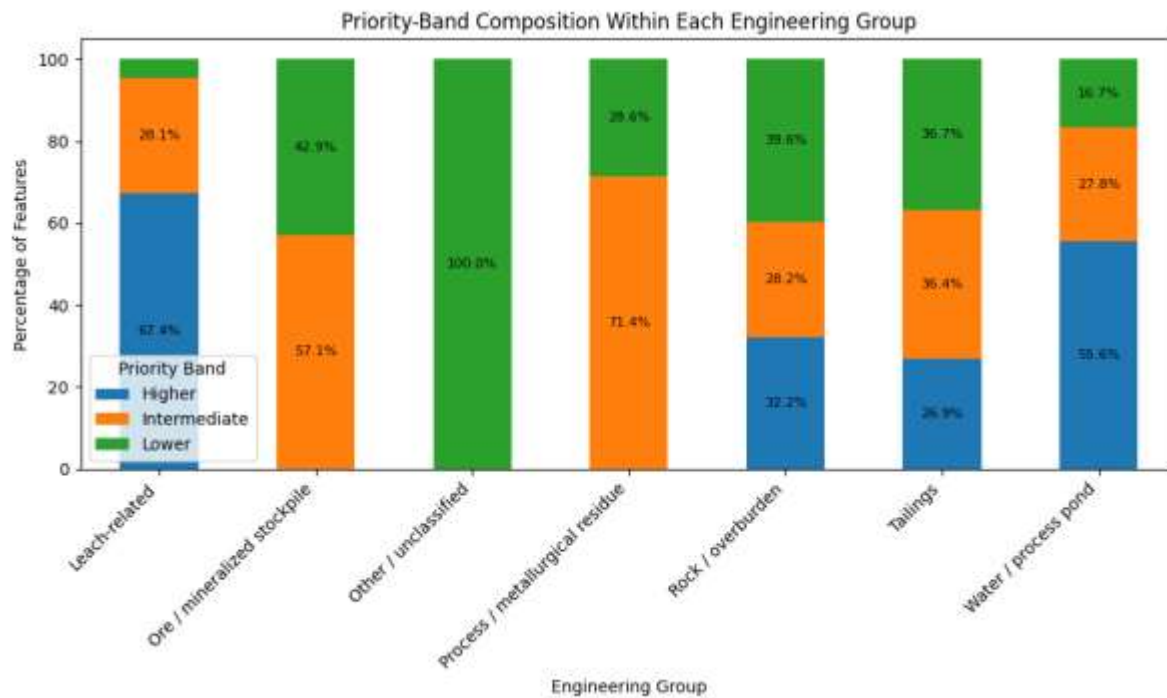


Figure 9.2. Priority-Band Composition within Each Engineering Group

9.3. Investigation-Profile Distribution

The resulting Investigation Profiles were distributed across the Priority Bands as follows:

Table 9.3. Investigation-Profile Distribution

Investigation Profile	Higher	Intermediate	Lower
P1	183	209	239
P2	60	25	4
P3	10	5	3
P4	0	10	4
P5	0	4	3
P6	0	0	1

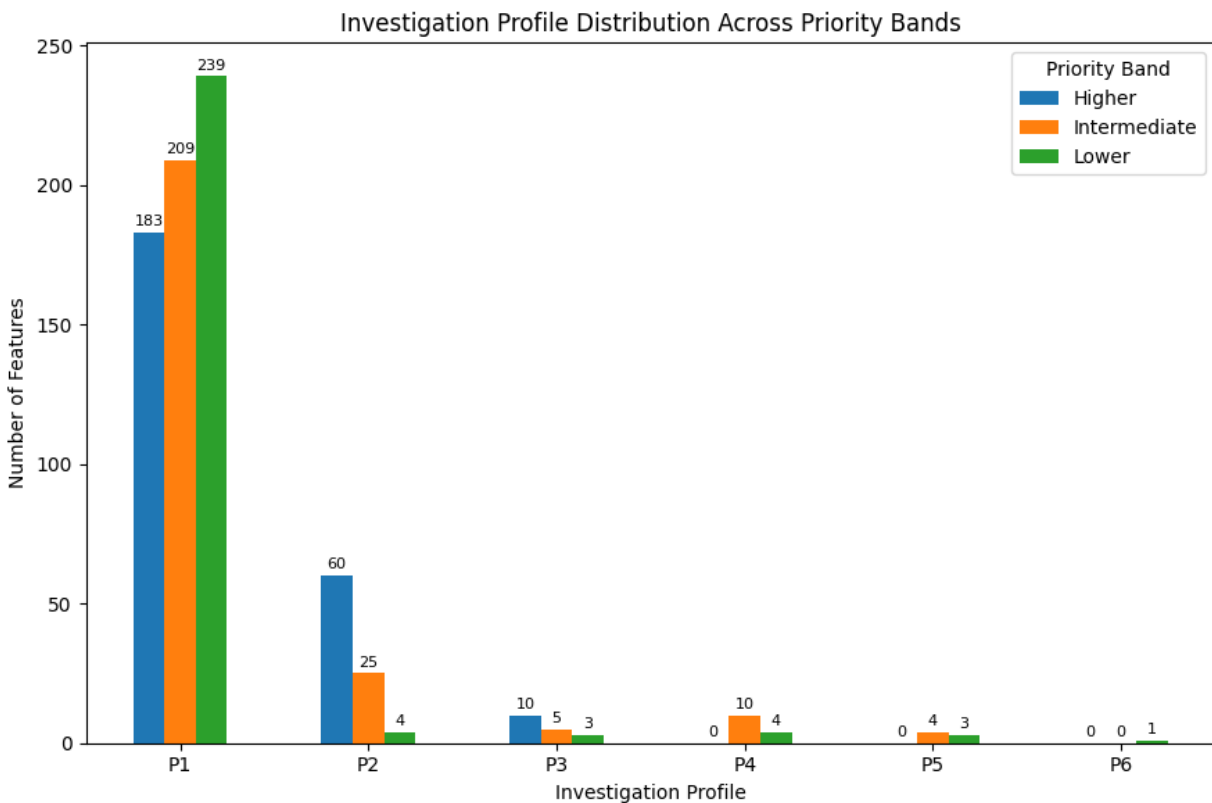


Figure 9.3. Investigation Profile Distribution Across Priority Bands

10. Engineering Interpretation

There are two complementary forms of output given by the framework. Based on the physical scale of features, Priority Band provides relative screening, while the investigation profiles give the information that supports the subsequent investigation based on four engineering domains.

These provide two distinct types of information and should not be combined to get a single numerical score.

The three priority bands, Higher, Intermediate, and Lower, contain 253, 253, and 254 features, respectively. This classification was generated using tertiles.

When observing the distribution of engineering groups in these priority bands, the influence of differences in volume distributions among waste types can be seen. For example, leach-related features can be seen in the higher priority band, while tailings and rock/overburden features can be seen across all three bands. This indicates that the screening process is entirely volume-dependent rather than bound to priority ranking based on waste types.

The investigation profiles interpret additional information that supports the subsequent investigation process. Features belonging to the P1 profile require prior investigation interest in material, geochemical, or geotechnical context, while water/containment is secondary. On the other hand, the features in the P2 profile require prior interest in geochemical and water/containment related investigation because of leach-related characteristics. For each profile, there is a specific engineering routing for further investigation based on their documented information.

A Higher-band feature should not be interpreted as a higher-risk or hazardous feature. Depending on the characteristics of the feature, some features in Lower-band may require prior investigation due to the site-specific conditions that are not considered in this preliminary screening framework. So, the framework must be treated as an initial screening tool rather than a final engineering decision.

11. Validation and Reproducibility

11.1. Data-to-Output Validation

There are 760 feature records with unique feature identifiers and 16 variables in the final engineering screening table. There were no missing values in fields that were considered for the final screening. This indicates that each valid record is represented once in the final output.

11.2. Priority-Band Validation

The calculated volume scores were used to recalculate the priority band assignments separately using two predefined thresholds to check validity. The resulting classifications have no mismatches with the final priority band classification.

This indicates the consistency of priority band classification with volume score and predefined tertile thresholds.

11.3. Investigation-Profile Validation

Using predefined P1-P6 profile mapping and based on the engineering-domain levels, the investigation profiles of each feature were reconstructed separately, resulting in no mismatches being detected.

Consistency throughout the feature type classification was proven, as each engineering group was associated with a single investigation profile.

11.4. Independence of the Priority Band from Non-Volume Variables

The framework was tested separately to check whether there is an influence from the other variables that are not involved in the screening process and whether they alter the priority band. The recalculated priority band using the volume thresholds was compared with the final classification. There were no mismatches identified.

The same validation method was carried out against the engineering profiles, engineering domain levels, management/context variables, commodity information, engineering groups, and feature-type classifications. But no influences were identified.

The final framework maintains the anticipated methodological separation, where the volume score gives the priority band classification, while feature classification determines the investigation profile, rather than allowing unintentional influence by the contextual or engineering profile variables on the primary screening order.

11.5. Boundary-Case Validation

Using the two priority band thresholds, boundary flags and boundary distances were recalculated separately, resulting in no mismatches.

There were 76 boundary flags, as their volume scores are within a 0.02 range of a band boundary. This is not considered an alternative classification rule, and these 76 features were retained within their original band.

11.6. Reproducibility

Using the two priority band thresholds, boundary flags and boundary distances were recalculated separately, resulting in no mismatches.

There were 76 boundary flags, as their volume scores are within a 0.02 range of a band boundary. This is not considered an alternative classification rule, and these 76 features were retained within their original classification. The framework was rerun separately using the same dataset and methodology. It produces a framework similar to the original framework but with predefined numerical tolerance.

The reproduced framework's fingerprint is different from the original framework's fingerprint, but the largest numerical difference was observed to be approximately 1.53×10^{-16} . This is because of

the floating-point numerical representation, and there is no effect on the final engineering outputs, priority band assignments, or profile classifications.

Therefore, with a predefined numerical tolerance, reproducibility can be evaluated in the final band.

12. Limitations

During the interpretation of the framework, several limitations were considered. The first limitation is that this framework was constructed based on their physical scale. The features were assigned to priority bands according to their volume score, which may cause a relatively small feature that belongs to the Lower band to require prior investigation based on its site-specific conditions.

During the construction of the investigation profiles using engineering domain levels, evidence based on the waste-feature-type level was considered. The physical, chemical, or geotechnical properties of each individual feature were not measured. This information was obtained using field and laboratory sampling by the USGS characterization protocols. This indicates there should be additional information beyond an inventory classification (Campbell et al., n.d.).

Grade, contained material, resource class, and some geological attributes were very useful information available in the inventory, but those are not complete for every feature. Therefore, these attributes were not considered as a final framework, resulting in a limited extent to address specific investigation urgencies such as economic value, resource recovery, or geological conditions.

This framework does not provide any measurement of physical measures such as slope stability, groundwater conditions, seepage, contaminants, or other site-specific performance, as it is based on desktop inventory information. These mine waste features have strong site-specific conditional dependence on their behavior (Vriens et al., 2020).

These methodologies are specific only to this study population and cannot be transferred to another mine-waste inventory without reassessment.

This framework can only be used as a transparent preliminary screening method to organize investigation efforts, rather than for a specific engineering assessment.

13. Conclusion

This study used the 2026 USGS National Mine Waste Inventory to develop a transparent desktop screening framework to prioritize mine-waste features for further investigation. 760 mine-waste features were selected based on their valid volume records for the screening. The primary screening variable was feature volume because, throughout the population, volume data were consistent.

Feature-level calculated volume was transferred to a logarithmic transformation and assigned a volume score between 0 and 1 using min-max normalization. Three priority bands were created based on the volume score: Higher, Intermediate, and Lower using tertile thresholds. Each band contains 253, 253, and 254 features, respectively.

Two outputs were generated from the framework. The second output, the engineering investigation relevance, is different from the screening order. The priority band prioritizes the features based on their physical scale, while investigation profiles interpret which engineering domains require prior consideration during subsequent investigation. Considering four main engineering domains, material, geochemical, geotechnical, and water/containment, six investigation profiles were established. These separate outputs avoid direct measurement of any specific investigation urgency, such as environmental risk, economic value, resource recovery potential, or engineering failure.

The output indicates that a simple inventory framework can reproduce a preliminary screening mechanism without any unsupported factors assigned to individual features. Investigation profiles provide additional information for subsequent investigation, as a generalized waste-type characterization rather than site-specific characterization.

Overall, this framework provides a manageable preliminary investigation priority classification, with transparency and reproducibility.

14. Future Improvement

Site-specific information such as slope geometry, groundwater conditions, material characteristics, contaminant information, and physical conditions can be used in future investigation priority classifications. Alternative investigation priority classification approaches can be taken by defining specific engineering purposes, such as environmental risk, economic value, or resource recovery, with relevant attributes.

15. References

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16. Appendices

Appendix A – Representative Screening Records

Representative records from the final engineering screening table:

Ftr_ID	Ftr_Type	Engineering_Group	Volume_Score	Priority_Band	Investigation_Profile	Material	Geochemical	Geotechnical	Water_Containment	Reclaimed	Superfund	Boundary_Distance	Boundary_Flag
UT000002-005	Mine dump	Rock / overburden	1	Higher	P1	2	2	2	1	No	Yes	0.339792412	FALSE
AZ000008-013	Leach pile - heap	Leach-related	0.977383611	Higher	P2	2	2	1	2	No	No	0.317176023	FALSE
AZ000005-002	Tailings - pond	Tailings	0.933993265	Higher	P1	2	2	2	1	No	No	0.273785677	FALSE
AZ000018-002	Mine dump	Rock / overburden	0.660207588	Intermediate	P1	2	2	2	1	Unknown	No	0	TRUE
NV000007-002	Leach pile - heap	Leach-related	0.460206382	Intermediate	P2	2	2	1	2	No	No	0.046586373	FALSE
OK000001-087	Tailings - undifferentiated	Tailings	0.41508044	Intermediate	P1	2	2	2	1	No	Yes	0.00146043	TRUE
NC000005-003	Overburden pile	Rock / overburden	0.414555131	Intermediate	P1	2	2	2	1	Unknown	No	0.000935122	TRUE
NC000024-001	Tailings - pond	Tailings	0.413620009	Lower	P1	2	2	2	1	Unknown	No	0	TRUE
OK000001-030	Tailings - undifferentiated	Tailings	0.413501063	Lower	P1	2	2	2	1	No	Yes	0.000118946	TRUE
NC000010-004	Ore stockpile/storage	Ore / mineralized stockpile	0.253988904	Lower	P5	2	1	1	0	Unknown	No	0.159631105	FALSE
NC000015-001	Mine dump	Rock / overburden	0.090427123	Lower	P1	2	2	2	1	Unknown	No	0.323192886	FALSE
KY000007-002	Tailings - pond	Tailings	0	Lower	P1	2	2	2	1	Unknown	No	0.413620009	FALSE

Appendix B – Reproducibility Information

B.1. Computational Environment

Python 3.10.4 was used for data processing, calculations, priority band assignments, engineering-profile assignments, validation, and generation of the final screening outputs.

Python libraries used for analysis:

- Pandas
- NumPy
- Pyogrio
- matplotlib

B.2. Input Dataset

The 2026 USGS National Mine Waste Inventory was used as the primary dataset, accessed from the USGS mine-waste geodatabase. For the analysis, feature resource, polygon and geology information from the inventory used as required.

For the final framework 760 valid volume records containing features were selected.

B.3. Processing Workflow

Final processing workflow was as follows.

1. Mine-waste features were identified using Ftr_ID.
2. The median Calc_Vol values were used to aggregate resource-level calculated volumes to the feature level for each Ftr_ID.
3. Missing and negative volume records were removed from the primary volume-based screening population.
4. Valid feature-level volume values were transformed into Log_{10} .
5. The Volume_Score was produced between 0 to 1 using min-max normalization.
6. One-third and two-thirds quantiles of Volume Score were used to divide the inventory into three bands, Lower, Intermediate and Higher.
7. Predefined four domain engineering matrix was used to assign engineering investigation profiles.
8. Reclaimed status, Superfund status, Commodity, Commodity Count, Boundary Distance, and Boundary Flag were considered as contextual variables.
9. The final framework outputs were validated against the defined methodological rules.